

Ashwin Amalaruban

Aviran Lab | University of California at Davis

Email: ashm@ucdavis.edu | Phone: +1 (530)-490-1492

Education

- **Ph.D. in Chemical Engineering** (*In progress*) | *University of California, Davis* Sep '23-Present
Cumulative GPA: 3.92/4
- **M.Tech in Chemical Engineering** | *Indian Institute of Technology, Bombay, India* July '19-July '21
Overall CPI: 9.35/10
Advisor: Dr. Jyoti Seth, co-advisor: Dr. Y.S. Mayya
Thesis: "Hydrodynamics of Fractal Aggregates"
- **B.Tech in Chemical Engineering** | *AC Tech, Anna University, India* Aug '14-Apr '18
Cumulative GPA: 8.19/10
Advisor: Dr. N. Nagendra Gandhi
Thesis: "Manufacture of Acetic Acid Using Zeolite"

Scholastic achievement

- 'Professor Kartic Khilar Prize (M.Tech)' for Outstanding Master's Dissertation, IIT Bombay 2021
- Secured **All India Rank 121 | 99.2 percentile** in GATE Chemical Engineering among **15537** candidates. 2019

Research and Technical Projects

RNA secondary structure prediction and thermodynamic parameter estimation | PhD research

Principal Investigator: Dr. Sharon Aviran, Collaborators: Dr. Dave Mathews, Dr. Max Ward Jul '24-Feb '26

- Evaluated thermodynamics-based RNA folding predictions against structure probing data to quantify where SHAPE-guided models improve over the standard Turner nearest-neighbor parameter set.
- Built a machine learning classifier to predict helix presence/absence using structure probing features and thermodynamic folding outputs.
- Developed MCMC based inference frameworks to refine nearest-neighbor thermodynamic parameters from experimental reactivity data.
- Extended parameter inference to modified nucleotides to quantify how chemical modifications reshape RNA folding stability and structural ensembles.

Hydrodynamics of Fractal Aggregates | Master's Thesis

Jan '20-Jul '21

Advisor: Prof. Jyoti Seth and Co-advisor: Dr. Y.S. Mayya

- Studied hydrodynamic mobility of fractal aerosol aggregates relevant to nuclear-reactor safety codes, where coagulated aerosols from accident scenarios can form highly irregular porous clusters.
- Generated aggregates using DLA, DLCA, and tunable CCA models, producing fractal dimensions spanning ~1.7–2.5.
- Simulated hydrodynamic behavior of 200–10,240 monomer aggregates using Stokesian Dynamics under creeping-flow conditions, calculating hydrodynamic radii and mobility coefficients.
- Developed a Python based multipole-expansion solver to compute 3D velocity fields around fractal geometries and quantify local flow perturbations.
- Reported the first Stokesian Dynamics based calculation of fluid penetration depth in fractal aggregates, resulting in a first-author *Langmuir* publication.

Manufacture of Acetic Acid using Zeolites | Bachelor's Project

Spring 2018

Advisor: Prof. N. Nagendra Gandhi, AC Tech, Chennai.

- Designed a 150 TPD acetic-acid production process using zeolite-based catalysis, including material and energy balances, process-flow development, and equipment sizing.
- Modeled and optimized multicomponent distillation performance in Aspen HYSYS to meet target product compositions.

Publication

Amalaruban, Ashwin, et al. "Relationship Between the Mobility of Aggregates and Fluid Penetration Depth Across a Range of Fractal Dimensions Using Stokesian Dynamics." *Langmuir* 38.11 (2022): 3422-3433.

<https://doi.org/10.1021/acs.langmuir.1c03180>

2022

Selected Technical and Course Projects

Molecular Dynamics of Asphaltene Adsorption at Water–Oil Interfaces

Mar 2026

Collaborative Project | Collaborators: Prof. Shirley Ahn and Prof. Manikantan

- Built GROMACS water–oil–water systems to investigate asphaltene adsorption at oil–water interfaces.
- Simulated hexane/water and decane/water systems with varying asphaltene loading to probe interfacial adsorption dynamics.
- Analyzed center-of-mass trajectories, density profiles, and adsorption fractions to quantify interfacial accumulation and molecular transport.

Binding stability of RNA-DNA interactions: CRISPR-Cas system

Fall 2024

BIM254-Statistical Methods in Genomics | Prof. Sharon Aviran | UC Davis

- Implemented CRISPROff and CRISPRspec energy-parameter models based on RNA–DNA, RNA–RNA, and DNA–DNA duplex free energies to predict off-target binding for Cas9 and Cas12a.
- Analyzed binding and unbinding behavior for Cas9 and Cas12a across target and off-target sequences.
- Benchmarked predictions on 4 target and 16 off-target sequences from a CIRCLE-seq dataset.

Catalytic Enzymes are Active Matter

Spring 2024

Preliminary Exam

- Reviewed experimental evidence for catalytic anti-chemotaxis and reaction enhanced enzyme diffusion using FCS/STED and TIRF single-molecule imaging studies.
- Evaluated substrate concentration dependent diffusivity trends, Michaelis-Menten type saturation arguments, and experimental limitations including FCS artifacts, photobleaching, and 2D projection effects.

Adaptive Stokesian Dynamics for interparticle interactions

Spring 2024

MAT226C - Numerical Methods: Ordinary Differential Equations | Prof. Robert Guy | UC Davis

- Developed and compared explicit Euler, fourth-order Runge–Kutta, Bogacki–Shampine RK23, and RKF45 integrators in Python for Stokesian Dynamics simulations of colloidal clusters.
- Showed that adaptive RKF45 reduced particle-overlap corrections by over 50% relative to fixed-step RK4 and required roughly one-third fewer time steps in high-force regimes.

Strategies for solving full-rank, ill-conditioned Linear systems and Eigenvalue problems

Winter 2024

MAT226B - Numerical Methods: Large-Scale Matrix Computations | Prof. Thomas Strohmer | UC Davis

- Compared iterative solvers, including GMRES and conjugate-gradient variants, for full-rank ill-conditioned linear systems.
- Evaluated QR, Lanczos, and Chebyshev approaches for extreme eigenvalue extraction in large-scale matrix problems.

Doi-Ohta model

Spring 2021

Rheology of complex fluids | Prof. Jyoti Seth | IIT Bombay

- Reviewed the model for predicting the rheology of immiscible polymer blends.
- Studied model corrections and recent extensions developed to improve agreement in complex-fluid systems.

Teaching and Mentorship Experience

Teaching Assistant

1. ECH 1Y: Coffee Lab | *Prof. Tonya Kuhl and Prof. William Ristenpart | UC Davis.* Winter '26, Spring '26
2. ECH143: Mass Transfer | *Prof. Brian Higgins | UC Davis* Spring 2024
3. CL-254: Process Fluid Mechanics | *Prof. Chandra Venkatraman | IIT Bombay* Spring 2021
4. CL-249: Computational Methods Lab | *Prof. Jyoti Seth | IIT Bombay* Autumn 2020

5. CL-202: Data Analysis and Interpretation | *Prof. Sharad Bhartiya* | *IIT Bombay*

Spring 2020

Mentorship

BIM189C: BioIntegrated Projects-Data Science | *Prof. Sharon Aviran* | *UC Davis*

Winter '25-Winter '26

- Mentored a team of undergraduate students on a project involving the classification of RNA secondary structure helices using SHAPE data and RNA thermodynamic models.
- Led hands-on sessions on Linux shell scripting, high-performance computing (HPC), and effective debugging in PyCharm IDE.
- Delivered a presentation on **JAX** and **automatic differentiation**, demonstrating its importance.

Readership

ECH158BN: Process Design | *Prof. Jason White* | *UC Davis*

Winter 2024

Technical skills

- **Programming Languages:** Python, MATLAB.
- **Scientific computing & Data Analysis:** NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Seaborn.
- **Parallel Computing & HPC:** multiprocessing, mpi4py, Linux.
- **Bioinformatics:** RNAstructure, ViennaRNA.
- **Molecular Dynamics:** GROMACS (introductory).
- **Software:** L^AT_EX, Aspen HYSYS/Plus.

Industry experience

Assistant Manager

Coke Oven Battery 9&10 | Steel Authority India Limited (SAIL) | Bhilai Steel Plant

Aug '21-Aug '23

Internship

Petrokens Engineering and Services Private Limited | Chennai

Feb '18-Mar '18

Extracurricular activities

Social	Received Rajyapuraskar Award from the Governor of Tamil Nadu.	2010
	CHMS GSO representative for the academic year 2023-2024.	2023-24
Sports	Member of National Sports Organisation Anna University Chennai	2016
Others	Secured 1st position in 'Arcade' Fashion Show competition IIT Bombay	2019